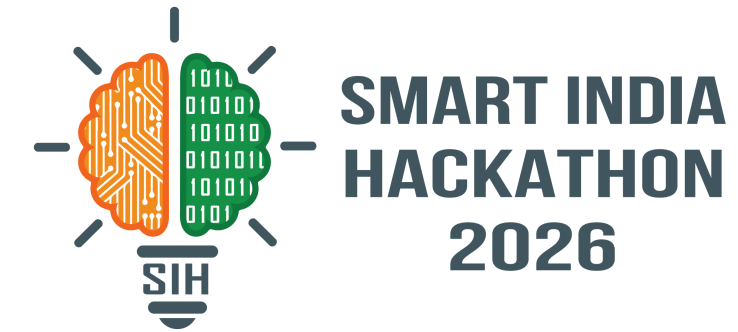


SMART INDIA HACKATHON 2026



- **Problem Statement Title:**

KISAN-DWAAR : An AI-Driven Platform for Farmer–Buyer Direct Connectivity, Equitable Price Discovery and Integrated Logistics

- **PS Category:** **Software**

- **Team Name:** **Macros**

- **Team Members Names:** **Akshat Jain**
Ashwani Visen
Disha Sirohi
Eindri Garg
Saksham Agarwal
Saksham Chaudhary



⚠ Understanding The Problem & Proposed Solution

Eliminating rural market exploitation through AI voice automation and government oversight.

❌ The Problem

- **Middlemen Exploitation:** Farmers receive lower prices while intermediaries capture most profits.
- **Limited Buyer Options:** Farmers are restricted to nearby traders despite better-paying buyers existing elsewhere.
- **Small Farmer Exclusion:** Low individual production cannot satisfy bulk commercial orders.
- **Transportation Barrier:** Many farmers don't own vehicles or can't find affordable transport at harvest time.
- **Poor Market Visibility:** Farmers cannot compare offers from multiple buyers in one place.
- **Digital Divide:** Most existing platforms require smartphones and internet access.

✅ Our Solution

- **Unified Buyer Marketplace:** Multiple verified buyers compete with different price offers.
- **Fair Price Recommendation:** AI suggests prices using mandi rates, MSP & demand.
- **Farmer Aggregation:** Combines nearby small to big scale farmers for bulk orders.
- **Community Logistics:** Matches nearby transporters and FPO vehicles.
- **Government-Verified Transactions:** Officials verify deliveries and payments.
- **Inclusive Access:** Works via IVR, SMS, and smartphone/web.

📞 Voice-First Innovation

100% Keypad Phone Inclusive: No internet or smartphone app required for rural farmers.

📞 Automated Farmer IVR Call:

"Namaste! Your wheat (500 kg) matches an order at ₹24.5/kg. Press 1 to Accept, Press 2 to Reject."

🚚 Automated Transporter IVR:

"Pickup 1000 kg load from Village A to Ghaziabad. Enter your rate quote using your keypad."

Technical Architecture Layers

Data Acquisition Layer: Aadhaar verification API, Mandi live price feeds, GPS location tracking, Telephony IVR triggers.

AI & ML Processing Layer: Fair Price Prediction (Regressors on historical Mandi + MSP data), Multi-Farmer Pooling Algorithm, Route Optimization Engine.

Voice Engine (Telecom Layer): Automated Speech Recognition (ASR) & DTMF keypad response processing in regional languages.

Decision & Governance Layer: Automated contract locking, reverse transport bidding engine, Government Official Oversight Portal.

</> Tech Stack & Infrastructure

AI / ML Core

PyTorch, Scikit-Learn, Pandas, Route Optimization Solvers, Price Regression.



Voice & Telephony

Twilio / Exotel IVR API, WebRTC, Multi-Lingual Speech Synthesis (TTS/STT).



Front-End & Portal

Next.js, Tailwind CSS, PWA for offline-first web portal, Admin Dashboard.



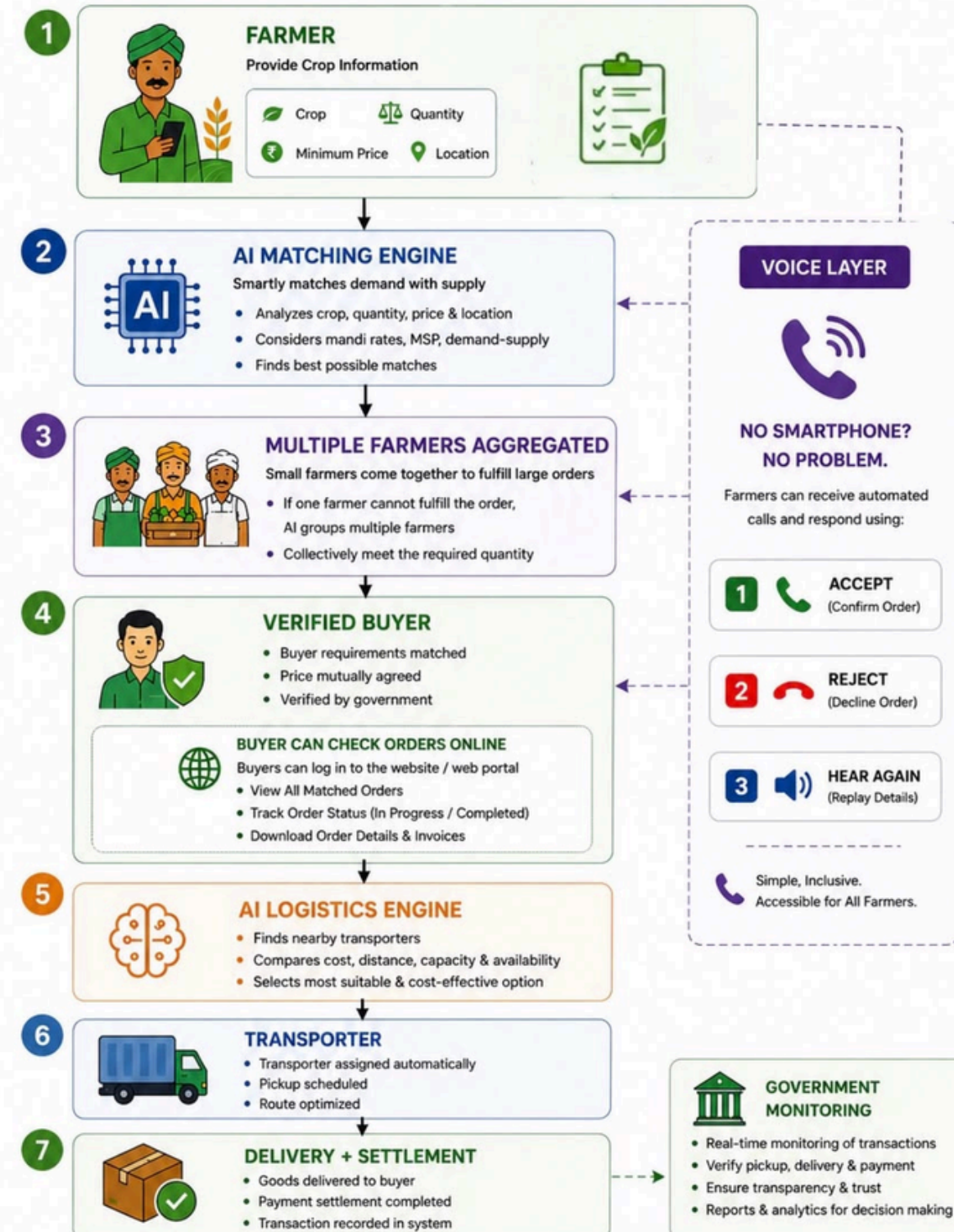
Backend & Security

FastAPI / Node.js, PostgreSQL, Redis, Aadhaar e-KYC Verification Gateway.



HOW IT SOLVES THE PROBLEM

An AI-Powered Government-Assisted Agricultural Procurement & Logistics Network



✔ Implementation Feasibility

- **Low Infrastructure Cost:** Utilizes existing mobile telecom networks without internet dependency for farmers.
- **Scalable Cloud Backbone:** Microservice API scales seamlessly across states and Mandi networks.
- **Government Office Network:** Leverages existing Gram Panchayat and Mandi officials for user onboarding.

🔄 Before vs After KisanDwaar

Before KisanDwaar

- Middlemen capture 30-40% margin.
- Farmers sell locally below MSP.
- Days lost searching for trucks.

After KisanDwaar

- 0% middleman commission.
- AI fair price matched to MSP & Mandi.
- Automated IVR truck system
- 100% keypad voice accessibility.

? "WHAT IFs...?" & Mitigation Strategies

WHAT IF farmer lacks smartphone?

System makes automated IVR voice calls. Farmer responds using keypad numbers (Press 1 to confirm).

WHAT IF transporters overcharge?

A system-generated base transport rate is shared with eligible transporters, who submit their quotes. The lowest eligible quote is automatically selected and locked for the deal.

WHAT IF yield is too small?

Multi-Farmer Aggregation Engine pools 3-5 neighboring smallholders to fulfill large 1000kg+ buyer requests.

WHAT IF payment fraud occurs?

Government verification officers audit delivery confirmation before escrow payment release to farmers.





Impacts and Benefits

Transforming rural livelihoods with measurable financial growth and transparent trade.



For Farmers

- **25-35% Income Surge:** Complete removal of middleman commission cuts.
- **Equal Market Entry:** Smallholders participate in large commercial orders.
- **Zero Digital Barrier:** Voice call access works on ₹1,000 keypad phones.

For Buyers & Logistics

- **Multiple order placing medium:** Buyer can use website and IVR system to place order.
- **Automated Logistics:** Transporters get steady job dispatches via IVR.
- **Reliable Quality:** Verified farmer network with transparent delivery tracking.

For Government

- **Total Transparency:** End-to-end monitoring from pickup to payment.
- **Price Stability:** AI Mandi price alignment prevents artificial hoarding.
- **Policy Intelligence:** Real-time crop yield & demand analytics dashboard.

Estimated Operational Improvement Impact (%)



Comparative Matrix & Technical References

Benchmarking KisanDwar against traditional and app-only agricultural solutions.

Feature / Aspect	KisanDwar (Our Proposed Solution)	Existing Conventional Solutions (APMC/Apps)
Accessibility	Voice-First IVR calls (Keypad phones) + Web/PWA	Smartphone applications only (Requires high internet & literacy)
Small Farmer Support	Multi-Farmer Aggregation Engine pools small yields	Individual listings only (Small farmers rejected for bulk orders)
Pricing Mechanism	AI Fair Price Engine (Mandi + MSP + Demand trends)	Manual bargaining or middleman dictated non-transparent pricing
Logistics Allocation	Competitive Transport Allocation for local transporters	Unorganized local transport search with fluctuating fees
Government Supervision	Official audit logging for verification & fraud protection	Minimal government oversight or completely unverified private trades

Research Standards, Data Sources & Telephony APIs

Agmarknet Mandi API: Live daily market price data integration across pan-India APMC Mandis.
<https://agmarknet.gov.in/>

e-NAM Data Integration: National Agriculture Market standard protocols for crop grading and MSP indexing.

Telecom IVR Frameworks: Multi-lingual automated telephony via WebRTC & Twilio/Exotel IVR gateways.